

Seminar 6 (Homework)

Discrete probabilities

1. If X, Y are independent random variables, then

$$\text{var}(X \cdot Y) = \text{var}(X)\text{var}(Y) + E(X)^2\text{var}(Y) + E(Y)^2\text{var}(X).$$

- 2 For X, Y, Z random variable, prove the following equalities:

$$\begin{aligned}\text{var}(aX + bY) &= a^2\text{var}(X) + b^2\text{var}(Y) + 2ab \text{cov}(X, Y), \quad \forall a, b \in \mathbb{R}, \\ \text{cov}(X + Y, Z) &= \text{cov}(X, Z) + \text{cov}(Y, Z).\end{aligned}$$

3. Suppose $X_1, X_2, \dots, X_n$ are independent with $E(X_i) = \mu$ and $\text{var}(X_i) = \sigma^2$. Find $E((X_1 + X_2 + \dots + X_n)^2)$

4. A computer randomly generates passwords consisting only of letters and numbers, distinguishing between capital and lowercase letters. If a password has 8 characters, what is the probability that it will contain 2 lower case letters, 3 capital letters and 3 digits (their order does not count).

5. Suppose that four new computer models M_1, M_2, M_3, M_4 are being tested for their reliability. The probability that a model satisfy the latest market standards are $p_1 = 0.8$ for model M_1 , $p_1 = 0.7$ for model M_2 , $p_3 = 0.9$ for model M_3 and $p_4 = 0.6$ for model M_4 . Determine the probability p that at least three models match the profile.

6. It was found that the probability to log on to a computer from a remote terminal is 0.7. Let X denote the number of attempts that must be made to gain access to the computer.

- Find the probability function of X . What type of distribution does X have?
- Find the expected value of attempts needed to gain access to the computer.
- Find the probability that at most 4 attempts must be made to gain access to the computer
- Find the probability that at least 3 attempts must be made to gain access to the computer.

7. We are given two urns, the first containing 6 white balls and 4 black balls and the second 8 white balls and 6 black balls. We extract a ball from the first urn and introduce it into the second. Then we extract 4 balls from the second urn. Let X be the number of white balls from the 4 extracted from the second urn. Which is the expected value of X ?

8. The telephone lines serving an information office are all busy about 60% of the times. Which is the expected number of tries for succeeding a reservation?

9. From an urn containing 8 white balls and 5 black balls we extract 3 times a ball, reintroducing it back each time. If X denote the number of white extracted balls, determine $E(X)$.

10. A multiple choice examination has 25 questions, each with 5 possible answers, only one of which is correct. Suppose that one of the students who takes the examination answers each of the question with an independent random guess.

- Write the random variable representing the number of correct answers.
- Which is the expected number of correct answers?

11. The number Y of customers per day at a certain sales counter has been observed for a long period of time and found to have a mean of 20 customers, with a standard deviation of 2 customers. The probability distribution of Y is not known. What can be said about the probability that Y will be between 16 and 24 tomorrow?

12. Let X and Y be independent random variables, each having a binomial distribution with parameters n, p and m, p , respectively, $p \in (0, 1)$. Which is the probability distribution of the random variable $X + Y$?

13. Eight letters are randomly distributed into 3 mailboxes. Let X be the number of letters in the 1st mailbox. Find the probability function of X .

14. Let $X_1, \dots, X_n$ be independent and identically distributed random variables, with

$$X_i : \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix}, \quad i \in \mathbb{N}_n.$$

Find the probability distribution of $Y = \sum_{i=1}^n X_i$. What type of distribution does Y have?

15. Let $X_1, \dots, X_n$ be independent and identically distributed random variables, with a geometric distribution of parameter p

$$X_i : \begin{pmatrix} k \\ pq^{k-1} \end{pmatrix}_{k=1,2,\dots}.$$

Find the probability distribution of $Y = \sum_{i=1}^n X_i$. What type of distribution does Y have?

16. Suppose that the number of missprints per page in a book of 400 pages are independent and Poisson distributed with parameter λ . Determine the probability that a page contains at least 2 misprints.

17. In an office n different letters are placed randomly into n addressed envelopes. Let Z_n denote the random variable that shows the number of correct mailings. For each $k \in \mathbb{N}_n$, let X_k be the random variable defined by

$$X_k = \begin{cases} 1, & \text{if the } k\text{-th letter is placed correctly,} \\ 0, & \text{otherwise.} \end{cases}$$

Calculate: a) $E(X_k)$ and $\text{var}(X_k)$ for each $k \in \mathbb{N}_n$.

b) $E(Z_n)$ and $\text{var}(Z_n)$

Answers:

4. Use multinomial scheme: $\frac{8!}{2!3!3!} \left(\frac{26}{62}\right)^2 \left(\frac{10}{62}\right)^3 \left(\frac{26}{62}\right)^3$ **5.** Use Poisson's urns scheme.

The required probability is $p = 0.7428$ **6.** a) Geometric distribution; b) $10/7$ c)

0.9919 d) 0.09 **11.** Use Chebyshev inequality: $P(16 < Y < 24) \geq \frac{15}{16}$ **12.** $X + Y$

has a binomial distribution with parameters $m + n, p$. **13.** See Problem 8, Seminar

4: $P(X = i) = \frac{2^{8-i} \binom{8}{i}}{3^8}$, $i = \overline{0, 8}$. **14.** Binomial with parameters n, p . **15.** Negative

binomial distribution. **16.** The r.v. showing the nr of missprints on a page:

$X : \begin{pmatrix} k \\ \frac{\lambda^k}{k!} e^{-\lambda} \end{pmatrix}_{k=0,1,\dots}$ The probability that a page contains at least 2 misprints is

$\sum_{k=2}^{\infty} P(X = k) = 1 - e^{-\lambda} - \lambda e^{-\lambda}$. **17.** $P(X_k = 1) = \frac{(n-1)!}{n!} = \frac{1}{n}$, $E(X_k) = \frac{1}{n}$, $\text{var}(X_k) = \frac{n-1}{n^2}$. b) $Z_n = X_1 + \dots + X_n$, but the r.v. in this sum are not independent!

We need $\text{cov}(X_i, X_j) = \frac{1}{n^2(n-1)}$, so finally $\text{var}(Z_n) = 1$.